Answers and talk notes

Understand Absolute Value of Rational Numbers

What your student is learning

Tonight we use absolute value to find the TOTAL distance between two points — including when they sit on opposite sides of zero.

Answers

Ask your student to explain before you look.

1. **B.** 16. −16 is 16 units from 0, so |−16| = 16.
2. **C.** 13. |−5| = 5 and |8| = 8, so 5 + 8 = 13.
3. **A.** 13. Add the distances from 0: 4 + 9 = 13.
4. **D.** 7. Same side, so subtract the distances: 12 − 5 = 7.
5. **B.** 16. The diver crosses 0: 10 + 6 = 16 feet.
6. **A.** |−25| + |40|. 0 is between −25 and 40, so add: |−25| + |40| = 65.

Talk about it

Does 0 fall between the two numbers? Do you add or subtract the distances from 0?

Home mission

Pick a spot in the room as zero. Put a sock 5 steps to its left and a spoon 3 steps to its right. Ask your student how many steps lie between them. Do they add or subtract? Why?

When they are stuck, ask

- Does zero fall between these two numbers?
- How far is each one from zero?
- Can a distance ever be negative?

Try not to say

- Just subtract the two numbers, distance is always subtraction.
- −8 is greater than −3 because 8 is greater than 3.
- Absolute value means to make it negative.